

Physics of Complex Systems Homework 2

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Problem 1 (Convolution product (6P)). Let us consider the probability density of the gaps between Poisson-distributed events (the length x of silence between the clicks of a Geiger counter), which is defined by

$$p(x) = we^{-wx} \quad (x \geq 0, w > 0).$$

- (a) Compute the corresponding characteristic function (see lecture notes). (1P)
- (b) Compute the two-fold convolution product $(p * p)(x)$ and the three-fold convolution product $(p * p * p)(x)$. (2P)
- (c) Guess a general formula for the N -fold convolution product $p^{*N}(x)$ and prove it by recursion. (2P)
- (d) Fit a normal distribution with mean μ and width σ to the probability density computed in (c) for large N .

Hint: Recall the behavior of cumulants. (1P)

Proof. (a) The characteristic function is given by

$$\phi_X(t) = E[e^{itX}].$$

We simply compute this

$$\begin{aligned} \phi_X(t) &= E[e^{itX}] \\ &= \int_0^\infty e^{itx} we^{-wx} dx \\ &= w \int_0^\infty e^{(it-w)x} dx \end{aligned}$$

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$$\begin{aligned}
&= w \left[\frac{1}{it - w} e^{(it-w)x} \right]_0^\infty \\
&= \frac{w}{w - it}
\end{aligned}$$

(b) The convolution of two probability distributions is given by

$$(f * g)(z) = \int_{-\infty}^{\infty} f(z-t)g(t) \, dt.$$

Again, we substitute this in, using a step function $\theta(t)$ to ensure that the probability distribution of the Poisson distribution only has support on the positive real axis.

$$\begin{aligned}
(p * p)(z) &= \int_{-\infty}^{\infty} w e^{-w(z-t)} \theta(z-t) w e^{-wt} \theta(t) \, dt \\
&= w^2 \theta(z) \int_0^z e^{-wz} e^{wt} e^{-wt} \, dt \\
&= w^2 z e^{-wz} \theta(z)
\end{aligned}$$

We compute the threefold product similarly

$$\begin{aligned}
(p * p * p)(z) &= w^3 \int_{-\infty}^{\infty} e^{-w(z-t)} \theta(z-t) t e^{-wt} \theta(t) \, dt \\
&= w^3 \theta(z) \int_0^z e^{-wz} t \, dt \\
&= w^3 \frac{z^2}{2} e^{-wz}
\end{aligned}$$

(c) We conjecture that

$$p^{*N}(x) = w^N \frac{x^N}{N!} e^{-wx} \theta(x).$$

We prove this by induction. Suppose this is true for N . We will prove it is true for $N+1$. We compute this product directly

$$\begin{aligned}
p^{*(N+1)}(z) &= (p * p^{*N})(z) \\
&= \int_{-\infty}^{\infty} w e^{-w(z-t)} \theta(z-t) w^N \frac{t^N}{N!} e^{-wt} \theta(t) \, dt \\
&= w^{N+1} \theta(z) \int_0^z e^{-wz} \frac{t^N}{N!} \, dt \\
&= w^{N+1} \frac{t^{N+1}}{(N+1)!} e^{-wz} \theta(z)
\end{aligned}$$

Since the base cases with $N = 1$ and $N = 2$ have already been proven, this proves the theorem.

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Problem 2. Exercise 4.2: Harmonic oscillator

(6P)

Consider a harmonic oscillator with eigenstates

$$|n\rangle = |0\rangle, |1\rangle, |2\rangle, \dots$$

and eigenenergies

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right).$$

Let us assume that the eigenstates correspond to classical configurations and that the system evolves stochastically according to a continuous-time Markov process as follows:

$$n \rightarrow n + 1 \quad \text{with rate } \omega_{\uparrow}$$

$$n \rightarrow n - 1 \quad \text{with rate } \omega_{\downarrow}$$

(a) Calculate the Liouville operator $\mathcal{L}$. (1P)

(b) Determine the left and the right eigenvector to the eigenvalue zero. (2P)

(c) Normalize the right eigenvector obtained in (b) so that it can be interpreted as a probability distribution. Under which condition is the normalization possible? (2P)

(d) In a canonical ensemble the probability of finding the system at energy E_n is proportional to

$$\exp\left(-\frac{E_n}{k_B T}\right),$$

where T is the temperature and k_B is the Boltzmann constant. Express the temperature T in terms of the rates $\omega_{\uparrow}$ and $\omega_{\downarrow}$. (1P)

Proof. (a) Recall that the Liouvillian is defined through the master equation

$$\partial_t |P_t\rangle = -\mathcal{L} |P_t\rangle.$$

Since it is stated that we may assume the eigenstates correspond to classical configurations, there is no need to concern ourselves with possible superpositions. Since

the rate of jumping up a step is given by $\omega_{\uparrow}$, there must be a term $-|n+1\rangle\langle n|\omega_{\uparrow}$, and similarly for $\omega_{\downarrow}$. Summing this up over all n , we find that the Liouvillian is given by

$$\mathcal{L} = -\omega_{\uparrow}|1\rangle\langle 0| + \omega_{\uparrow}|0\rangle\langle 0| - \sum_{n=1}^{\infty} [\omega_{\uparrow}|n+1\rangle\langle n| + \omega_{\downarrow}|n-1\rangle\langle n| - (\omega_{\uparrow} + \omega_{\downarrow})|n\rangle\langle n|].$$

(b) For the left eigenvector, we require that

$$\begin{aligned} 0 &= \langle \psi | \mathcal{L} \\ &= -\omega_{\uparrow} \langle \psi | 1 \rangle \langle 0 | + \omega_{\uparrow} \langle \psi | 0 \rangle \langle 0 | \\ &\quad - \sum_{n=1}^{\infty} [\omega_{\uparrow} \langle \psi | n+1 \rangle \langle n | + \omega_{\downarrow} \langle \psi | n-1 \rangle \langle n | - (\omega_{\uparrow} + \omega_{\downarrow}) \langle \psi | n \rangle \langle n |] \\ &= -\omega_{\uparrow} \langle \psi | 1 \rangle \langle 0 | + \omega_{\uparrow} \langle \psi | 0 \rangle \langle 0 | \\ &\quad - \sum_{n=1}^{\infty} [\omega_{\uparrow} \langle \psi | n+1 \rangle + \omega_{\downarrow} \langle \psi | n-1 \rangle - (\omega_{\uparrow} + \omega_{\downarrow}) \langle \psi | n \rangle] \langle n | \end{aligned}$$

Demanding that all components of this vector vanish then demand that

$$\langle \psi | 1 \rangle - \langle \psi | 0 \rangle = 0$$

and

$$\omega_{\uparrow} \langle \psi | n+1 \rangle + \omega_{\downarrow} \langle \psi | n-1 \rangle = 0$$

for all n .

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